

# Chapter 1 What is Data Mining?

After completing this Chapter, you will be able to:

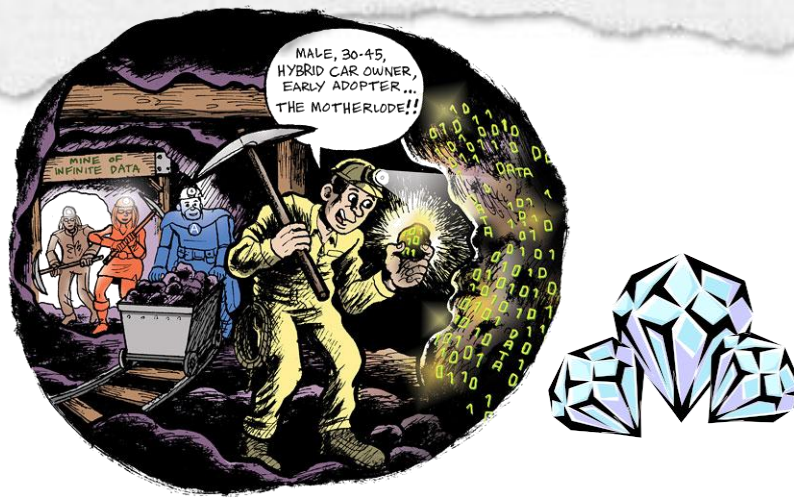
- Know what data mining (DM) is.
- Understand what DM emphasizes.
- Know the keys for successful DM.



# What is Data Mining?

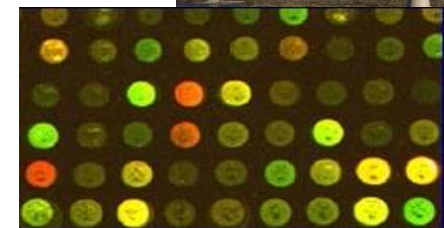
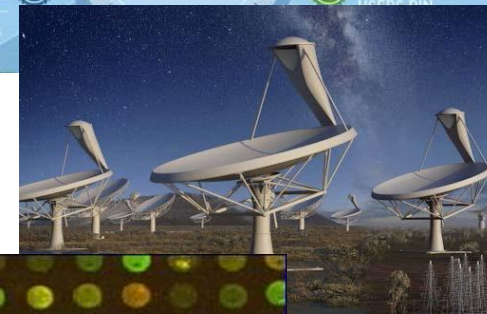
**Data Mining** is a process that uses a variety of **data exploration and analysis tools** to discover meaningful **hidden patterns and rules** (knowledge) in data.

**Knowledge** to be obtained can be discrimination, association, classification, clustering, trend, outliers, etc.



# Driving Forces

- **Lots of data** is being collected
  - Web data, e-commerce
  - Bank/credit card transactions
  - Purchases at supermarkets
  - Healthcare / Finance / Social media
  - telescopes scanning the skies
  - microarray generating gene expression data
- Computers have become **cheaper** and more **powerful**
  - Cheaper storage devices with higher capacity
  - Better database management system
- **Competitive pressure** is strong
  - Provide better and customized services (CRM)
- Traditional techniques **infeasible** for raw data



# 100K Wellness Project

## 100,000 People, 250 Biomarkers, and the Quest for Good Health

Hood et al. *BMC Medicine* 201513:4  
DOI 10.1186/s12916-014-0238-7



### COMMENTARY

### Open Access

## Integrating big data and actionable health coaching to optimize wellness

Leroy Hood<sup>1\*</sup>, Jennifer C Lovejoy<sup>2</sup> and Nathan D Price<sup>1</sup>

### Abstract

The Hundred Person Wellness Project (HPWP) is a 10-month pilot study of 100 'well' individuals where integrated data from whole-genome sequencing, gut microbiome, clinical laboratory tests and quantified self measures from each individual are used to provide actionable results for health coaching with the goal of optimizing wellness and minimizing disease. In a commentary in *BMC Medicine*, Diamandis argues that HPWP and similar projects will likely result in 'unnecessary and potential harmful over-testing'. We argue that this new approach will ultimately lead to lower costs, better healthcare, innovation and economic growth. The central points of the HPWP are: 1) it is focused on optimizing wellness through longitudinal data collection, integration and mining of individual data clouds, enabling development of predictive models of wellness and disease that will reveal actionable possibilities; and 2) by extending this study to 100,000 well people, we will establish multiparameter, quantifiable wellness metrics and identify markers for wellness to early disease transitions for most common diseases, which will ultimately allow earlier disease intervention, eventually transitioning the individual early on from a disease back to a wellness trajectory.

Please see related commentary: <http://dx.doi.org/10.1186/s12916-014-0239-6>.

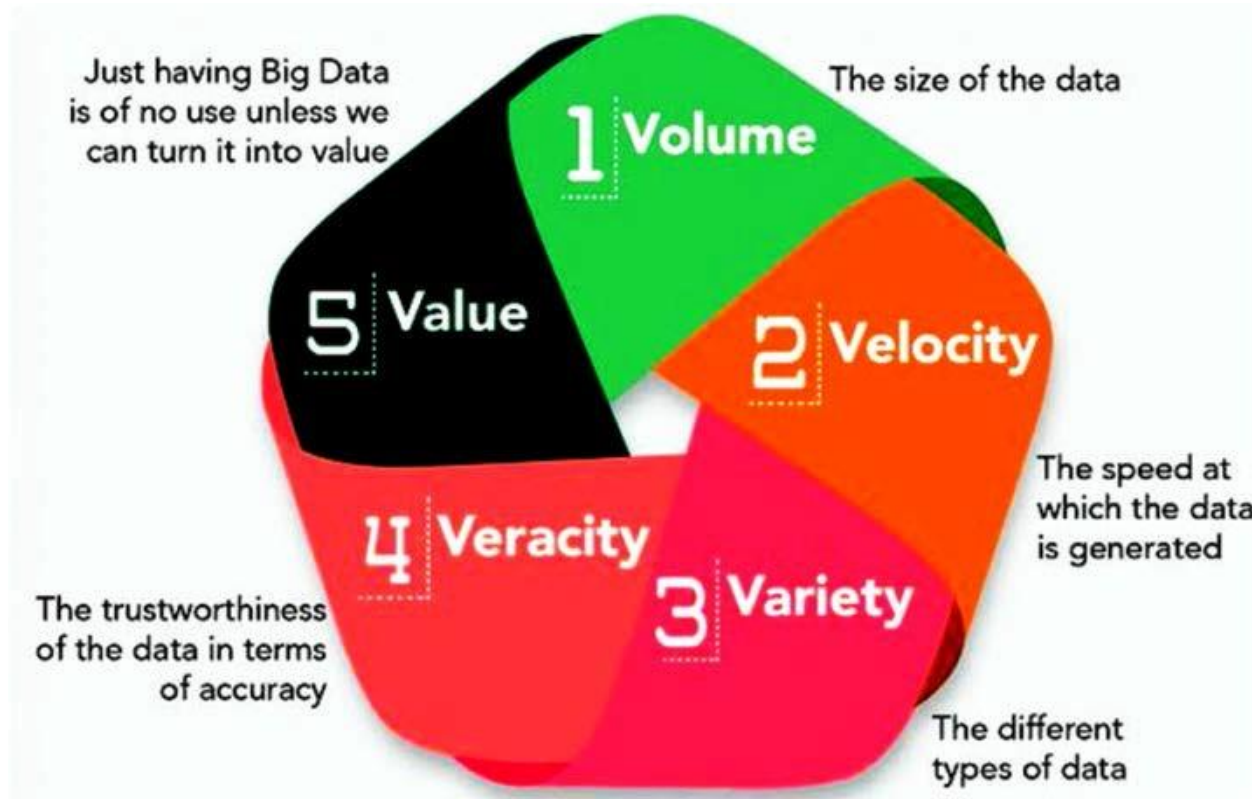
**Keywords:** Wellness, Personalized medicine, Whole-genome sequencing, Health behavior change, Actionable, P4 Medicine, Systems medicine, Gut microbiome

To hope to recruit 100,000 healthy individuals and collect data from whole-genome sequencing, gut microbiome, clinical laboratory tests and so on for 25 years, waiting for the healthy to turn sick.

To provide actionable results for health coaching with the goal of optimizing wellness and minimizing disease.

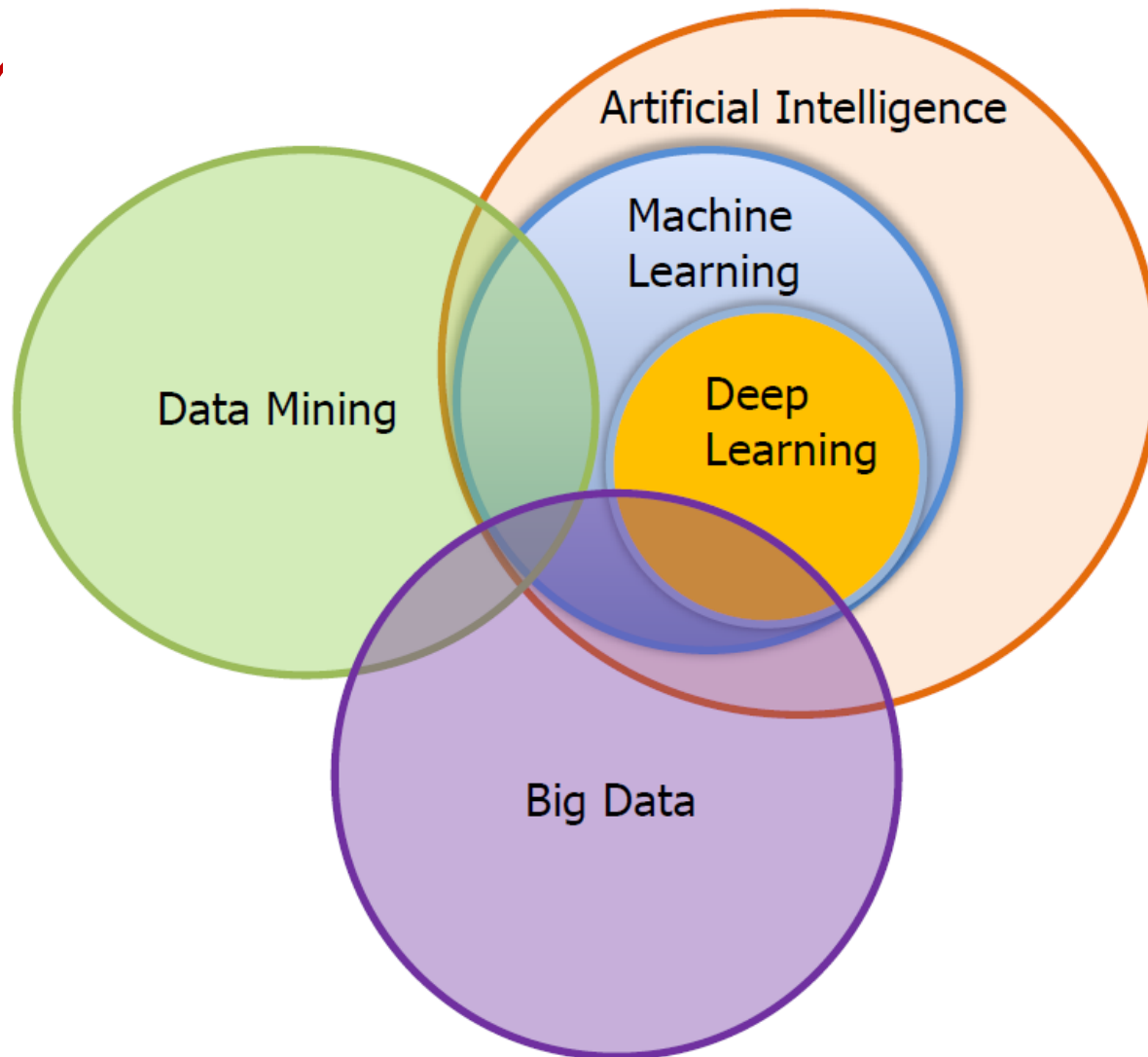
For an individual, researchers can monitor his body and then give him tips on improving his health.

# Big Data Characteristics



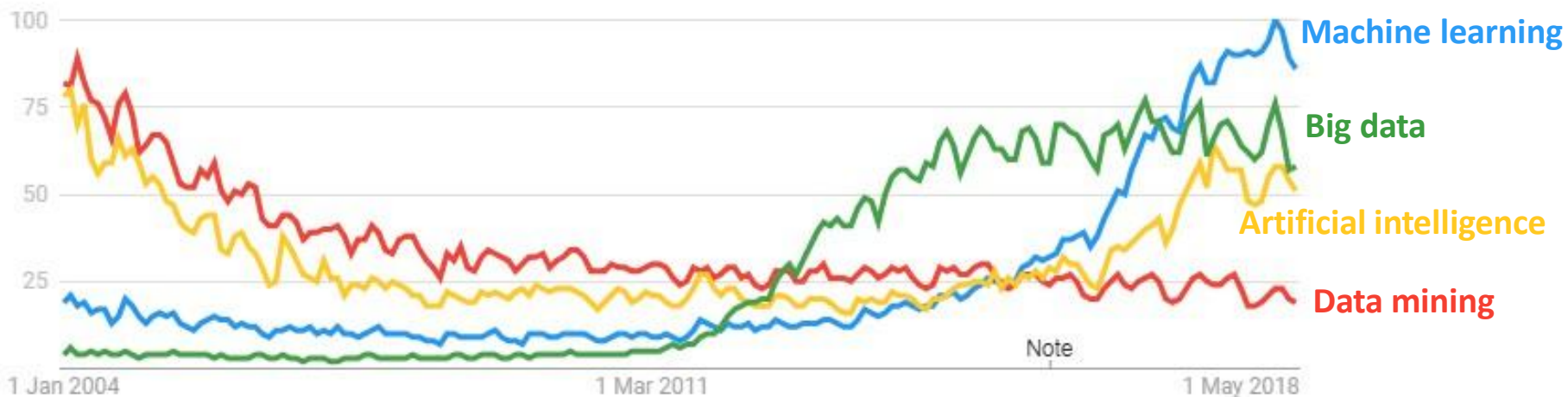
Source: Figure 3. The Five V's of Big Data from <http://pubs.sciepub.com/iteces/4/1/3/>

# Data Mining, AI, Big Data, Machine Learning and Deep Lear



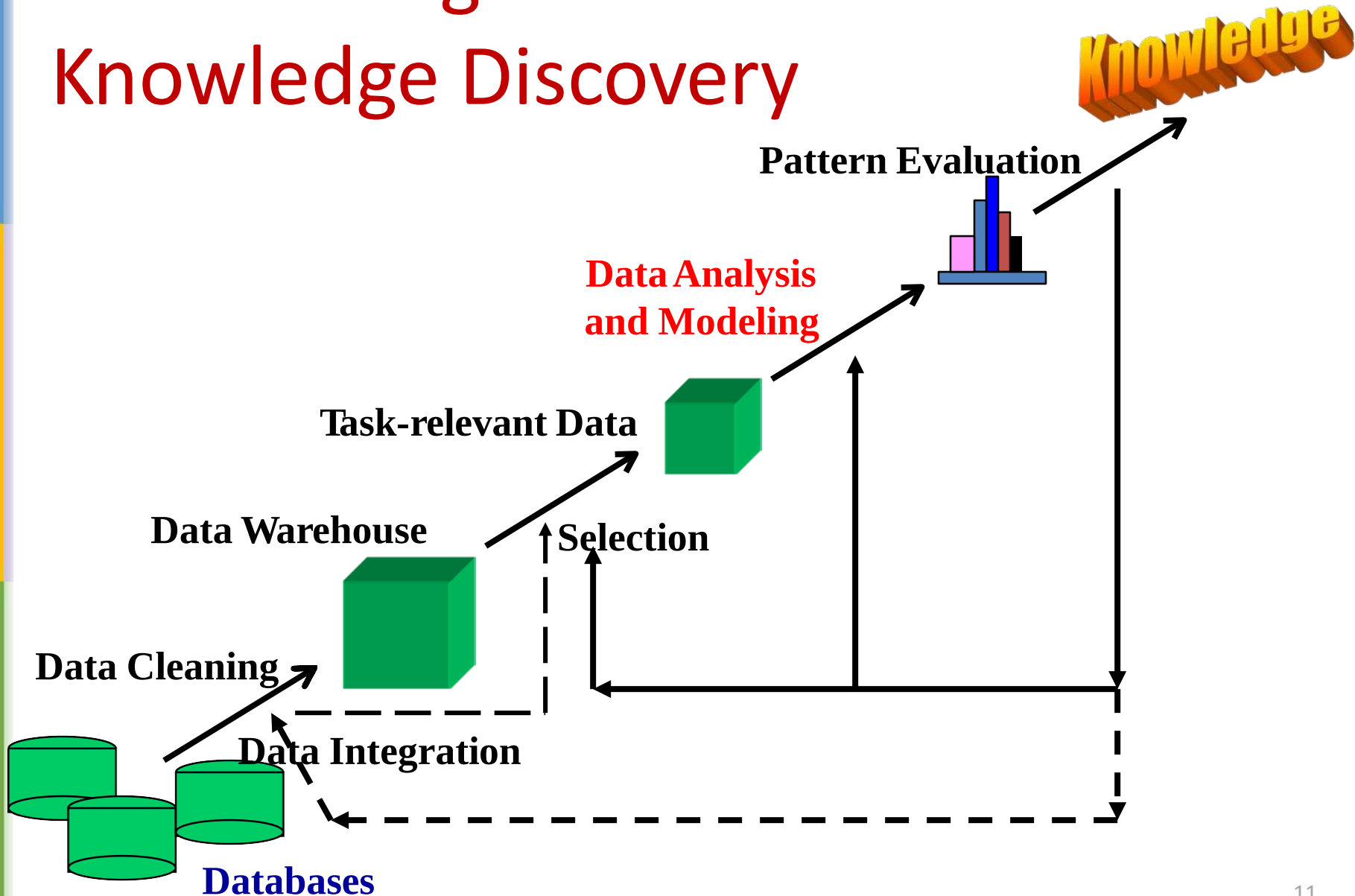
# Data Mining

- **Data mining:** a process of discovering patterns in large data sets
  - (Semi-)Automatic analysis of large data to identify interesting patterns
  - Using artificial intelligence, machine learning, **statistics** and databases
- Difference between DM and ML:
  - ML enables DM and DM employs ML



Google Search Trends, relative searches

# Data Mining Process: Knowledge Discovery



Source: Han, Kamber and Pei (2012)

# Data Mining Techniques

Two main tasks:

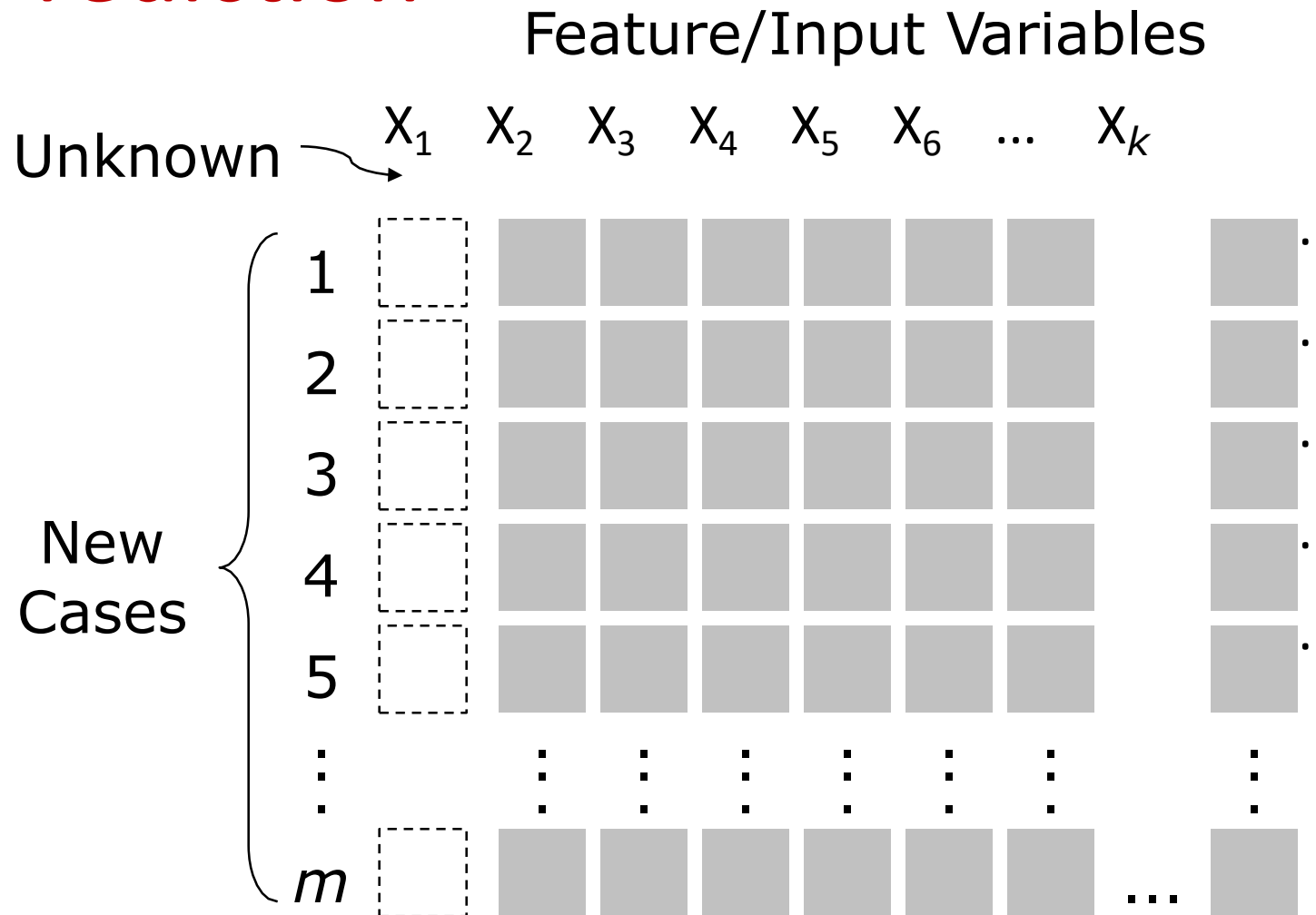
- Supervised data mining (Predictive Analytics)
  - Regression / logistic regression
  - Decision trees
  - Neural networks
  - Support vector machines
  - Discriminant analysis
  - Naive Bayes
  - K-nearest-neighbor models
  - Ensemble methods: bagging, random forests, boosting
- Unsupervised data mining (Descriptive Analytics)
  - Clustering; outlier analysis
  - Self-organizing maps
  - Principal components
  - Association rules / Sequence analysis
  - Link analysis

# Supervised Learning

|                     |     | Feature/Input Variables |       |       |       |       |       |     |       |
|---------------------|-----|-------------------------|-------|-------|-------|-------|-------|-----|-------|
| Label/Y<br>Response |     | $X_1$                   | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | ... | $X_k$ |
| Cases               | 1   |                         |       |       |       |       |       |     |       |
|                     | 2   |                         |       |       |       |       |       |     |       |
|                     | 3   |                         |       |       |       |       |       |     |       |
|                     | 4   |                         |       |       |       |       |       |     |       |
|                     | 5   |                         |       |       |       |       |       |     |       |
|                     | ⋮   | ⋮                       | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮   | ⋮     |
|                     | $n$ |                         |       |       |       |       |       | ... |       |

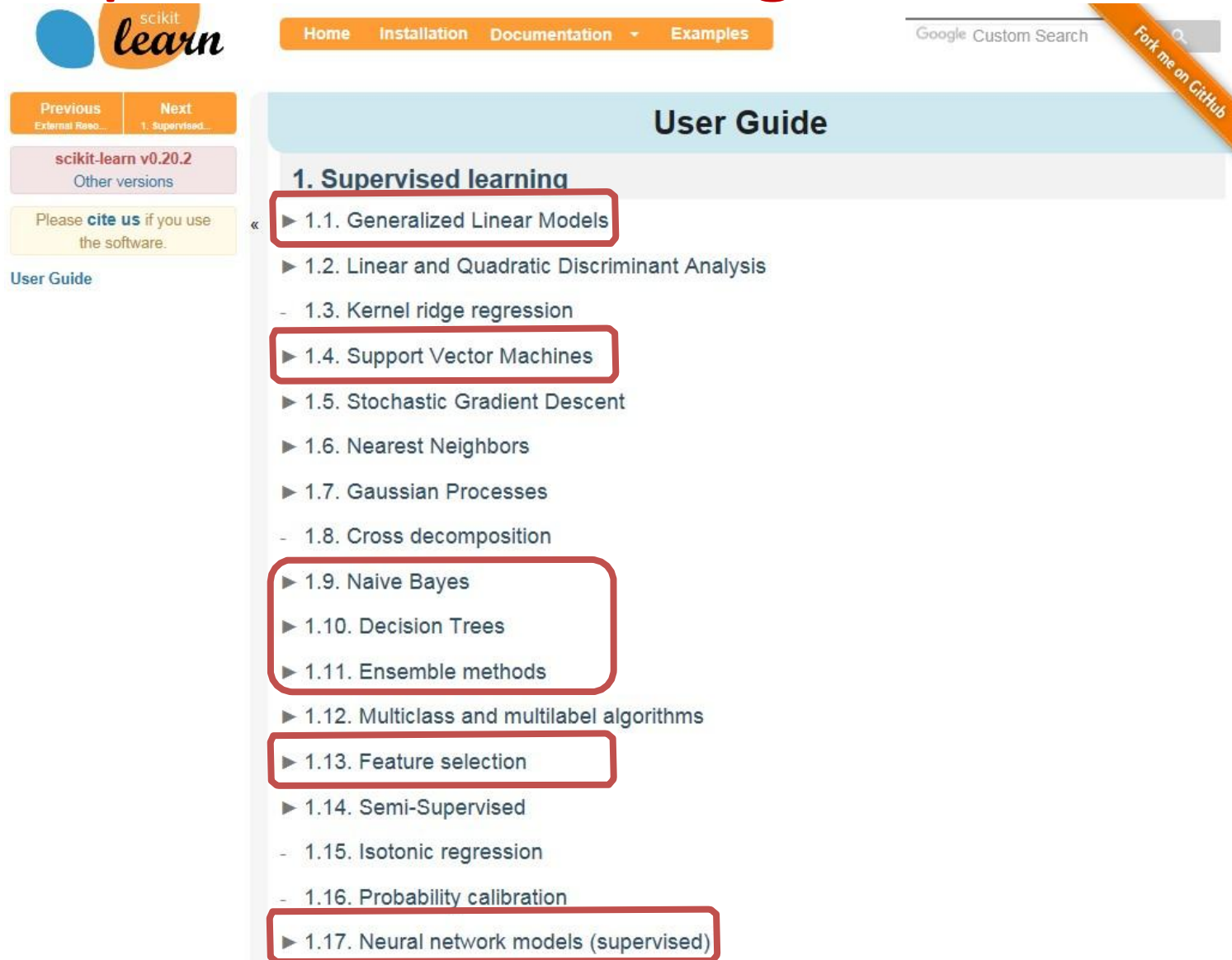
Model:  $Y = f(X_1, X_2, \dots, X_k; \theta)$

# Prediction



Fitted Model:  $\hat{Y} = f(X_1, X_2, \dots, X_k; \hat{\theta})$

# Supervised Learning in scikit-



The screenshot shows the scikit-learn website's User Guide for Supervised Learning. The page has a light blue header with the scikit-learn logo and navigation links: Home, Installation, Documentation, and Examples. A search bar with 'Google Custom Search' and a 'Fork me on GitHub' button are also present. The left sidebar contains links for 'Previous' (External Resources) and 'Next' (1. Supervised Learning), the current version 'scikit-learn v0.20.2', and a 'cite us' notice. The main content area is titled 'User Guide' and lists 17 topics under '1. Supervised learning'. The first 13 topics are highlighted with red boxes: 1.1. Generalized Linear Models, 1.2. Linear and Quadratic Discriminant Analysis, 1.3. Kernel ridge regression, 1.4. Support Vector Machines, 1.5. Stochastic Gradient Descent, 1.6. Nearest Neighbors, 1.7. Gaussian Processes, 1.8. Cross decomposition, 1.9. Naive Bayes, 1.10. Decision Trees, 1.11. Ensemble methods, 1.12. Multiclass and multilabel algorithms, and 1.13. Feature selection. The remaining four topics (1.14. Semi-Supervised, 1.15. Isotonic regression, 1.16. Probability calibration, and 1.17. Neural network models (supervised)) are not highlighted.

scikit-learn

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Previous External Resources Next 1. Supervised Learning

scikit-learn v0.20.2  
Other versions

Please [cite us](#) if you use the software.

User Guide

## 1. Supervised learning

- ▶ 1.1. Generalized Linear Models
- ▶ 1.2. Linear and Quadratic Discriminant Analysis
- 1.3. Kernel ridge regression
- ▶ 1.4. Support Vector Machines
- ▶ 1.5. Stochastic Gradient Descent
- ▶ 1.6. Nearest Neighbors
- ▶ 1.7. Gaussian Processes
- 1.8. Cross decomposition
- ▶ 1.9. Naive Bayes
- ▶ 1.10. Decision Trees
- ▶ 1.11. Ensemble methods
- ▶ 1.12. Multiclass and multilabel algorithms
- ▶ 1.13. Feature selection
- ▶ 1.14. Semi-Supervised
- 1.15. Isotonic regression
- 1.16. Probability calibration
- ▶ 1.17. Neural network models (supervised)

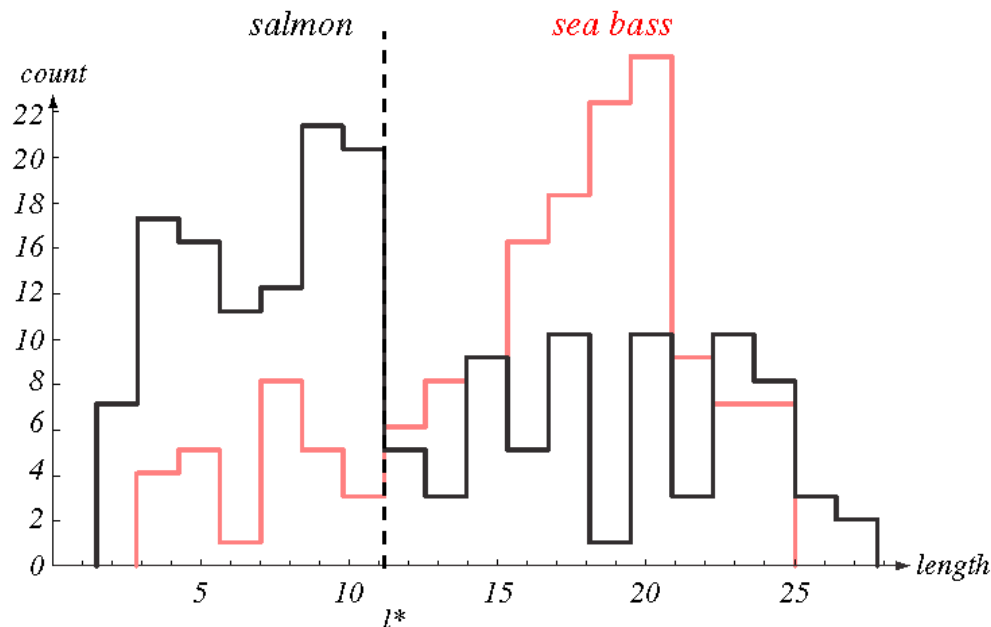
# Eg 1: Species Recognition

- **Goal:** Sorting incoming fish on a conveyor belt according to species using optical sensing
- **Two species of fish:**
  - sea bass
  - salmon
- Set up a camera and take some sample images to extract **features**
  - length, lightness, width, number and shape of fins, etc...
- **Steps:** capture image → isolate fish → take measurements → construct a **classifier**



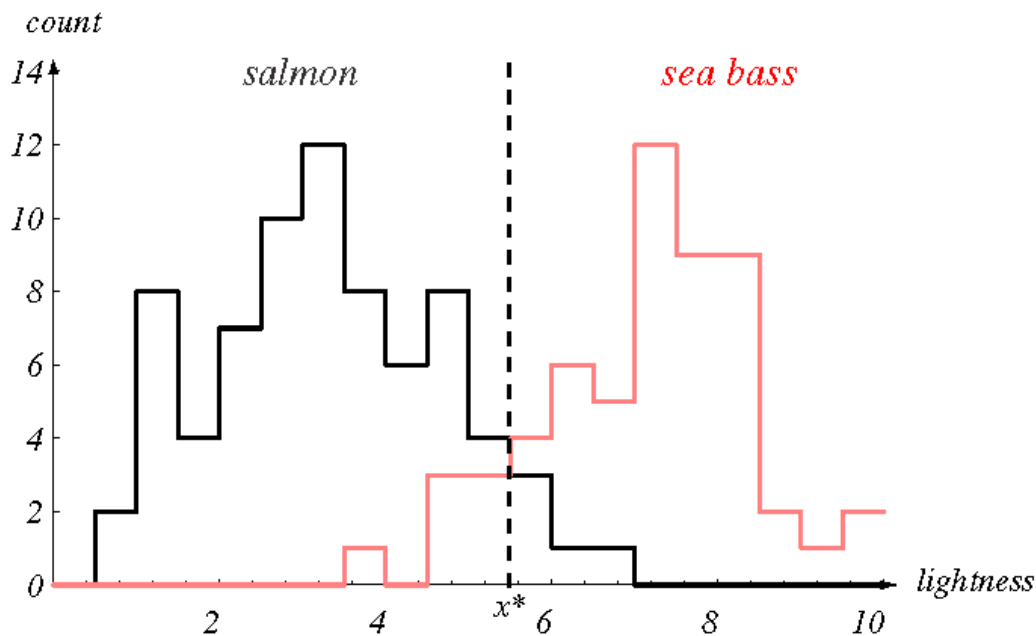
# Eg 1: Species Recognition

- A fisherman told you that a sea bass is generally longer than a salmon.
- Consider **length** as a feature for discrimination
- How can we choose this threshold?
- Is length a good feature?



# Eg 1: Species Recognition

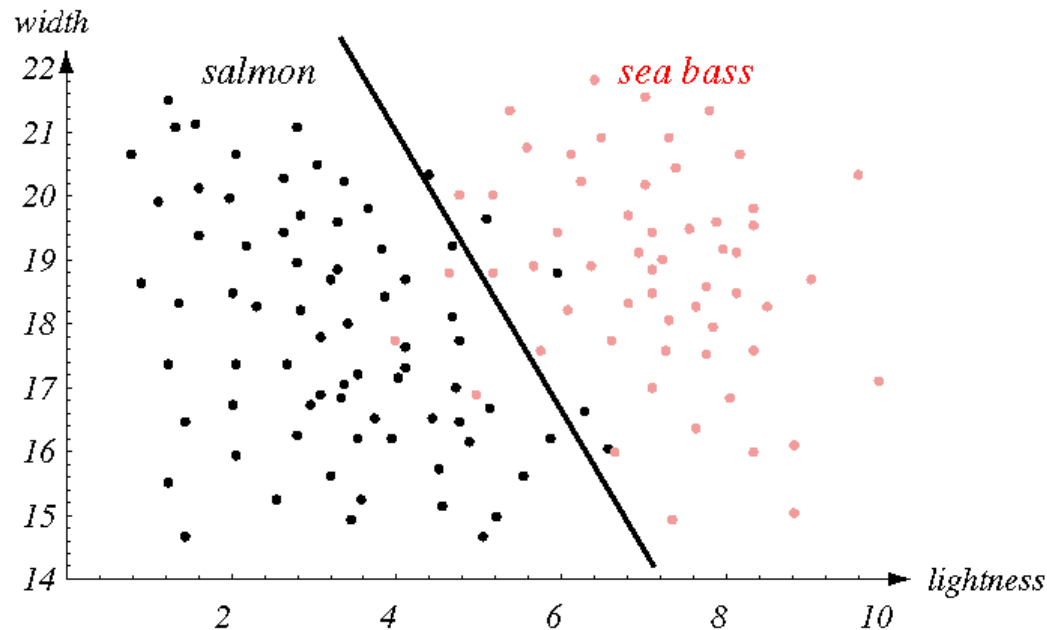
- Try another feature: lightness



- It looks easier to choose the threshold  $x^*$  but the classification is still not perfect.

# Eg 1: Species Recognition

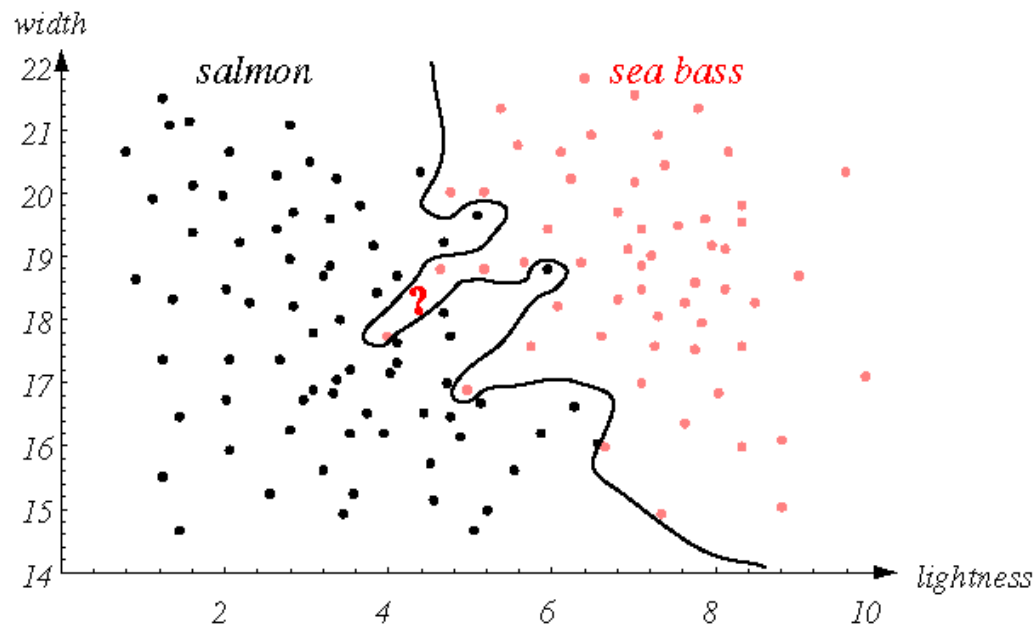
- Try multiple features: lightness and width



- We can draw a **decision boundary** to divide the **feature space** into two regions. How to determine the boundary?
- Does it look better than using only lightness?

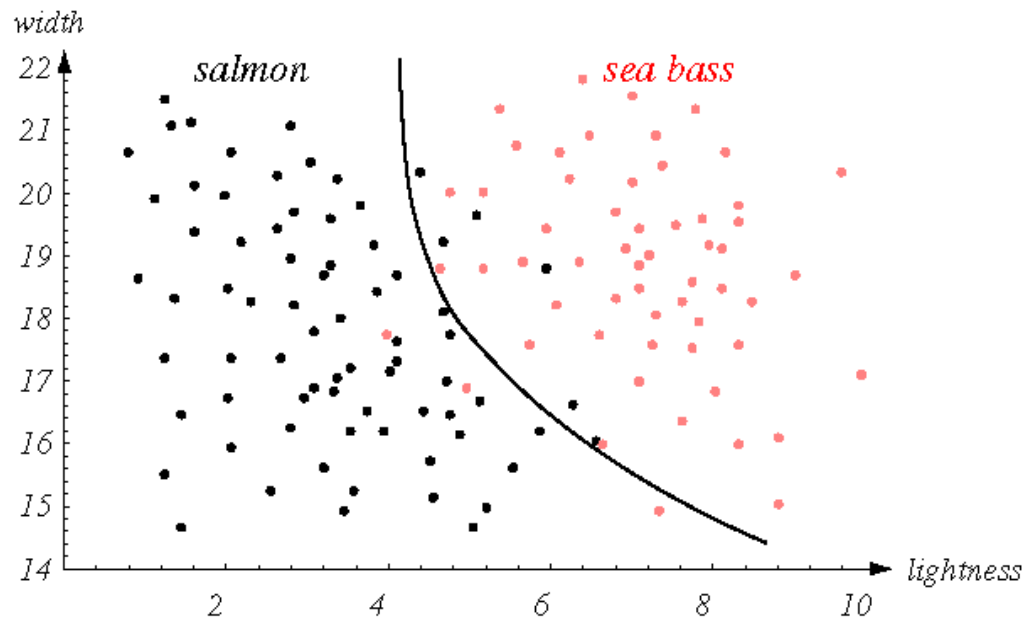
# Eg 1: Species Recognition

- Does adding more features help improve the results?
  - Avoid noisy/irrelevant features
  - Be careful about strong correlations with existing features
  - Avoid overfitting (issue of generalization)

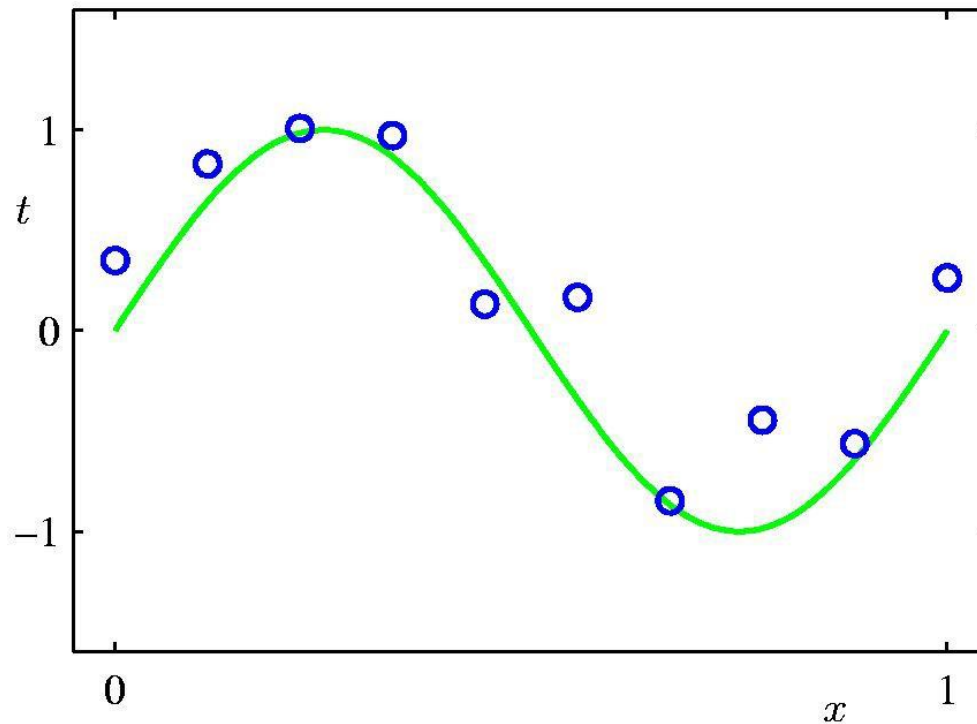


# Eg 1: Species Recognition

- A **tradeoff** between complexity of decision rules and their performance to unknown samples:

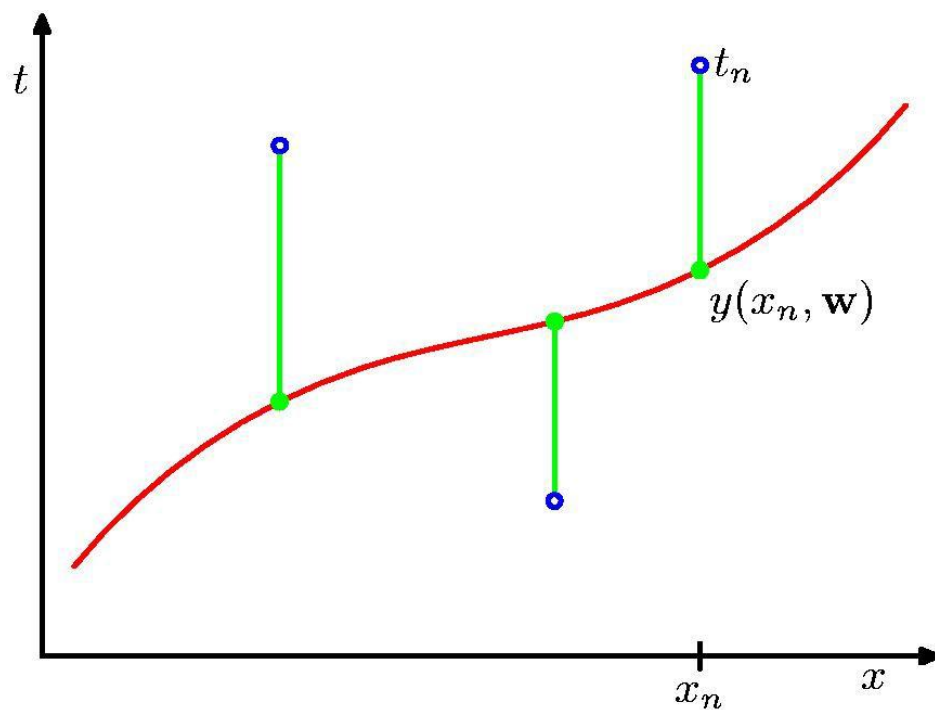


## Eg 2: Polynomial Curve Fitting



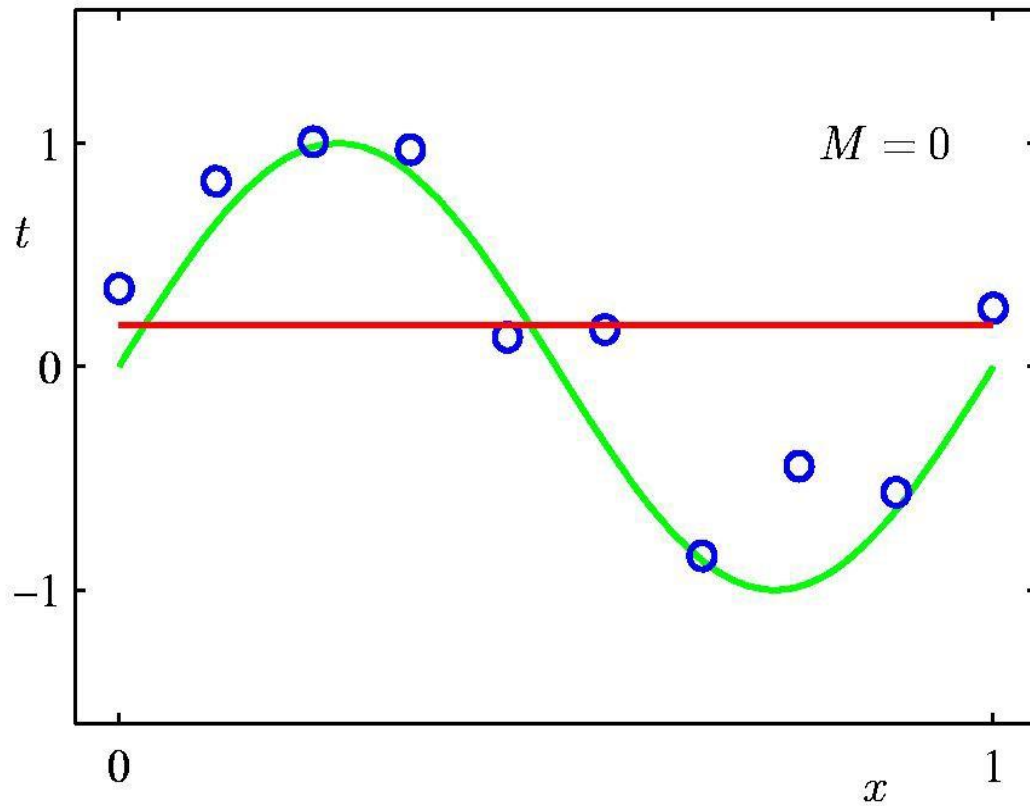
$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j$$

# Sum-of-Squares Error Function

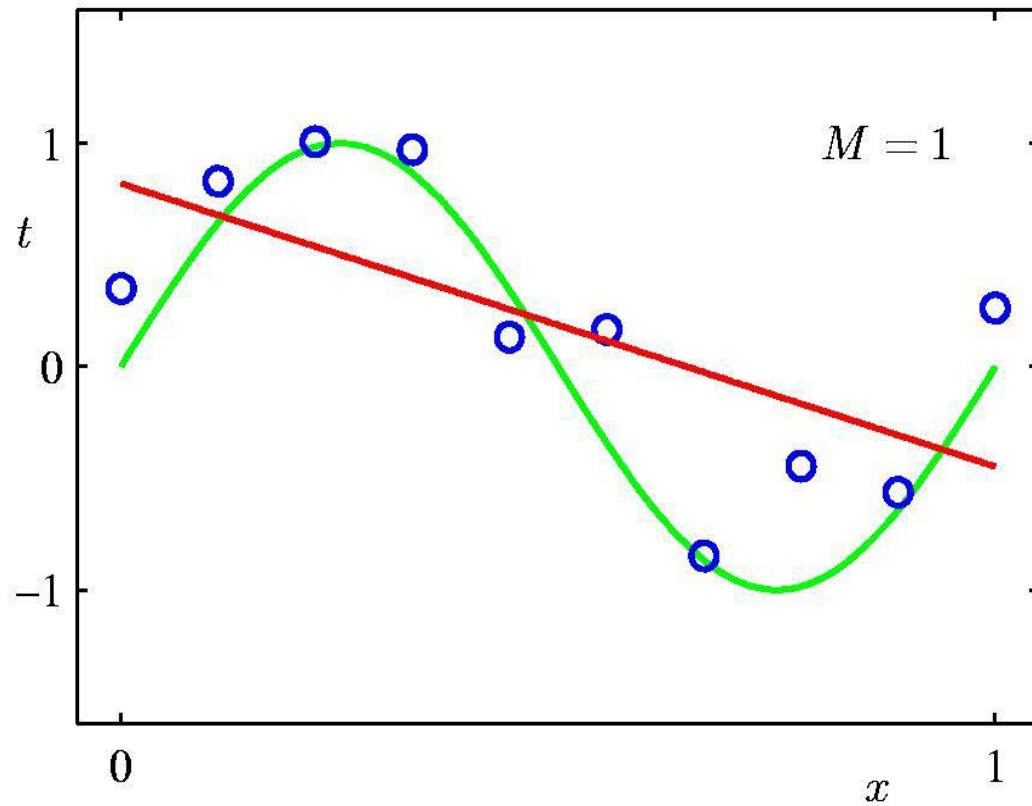


$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

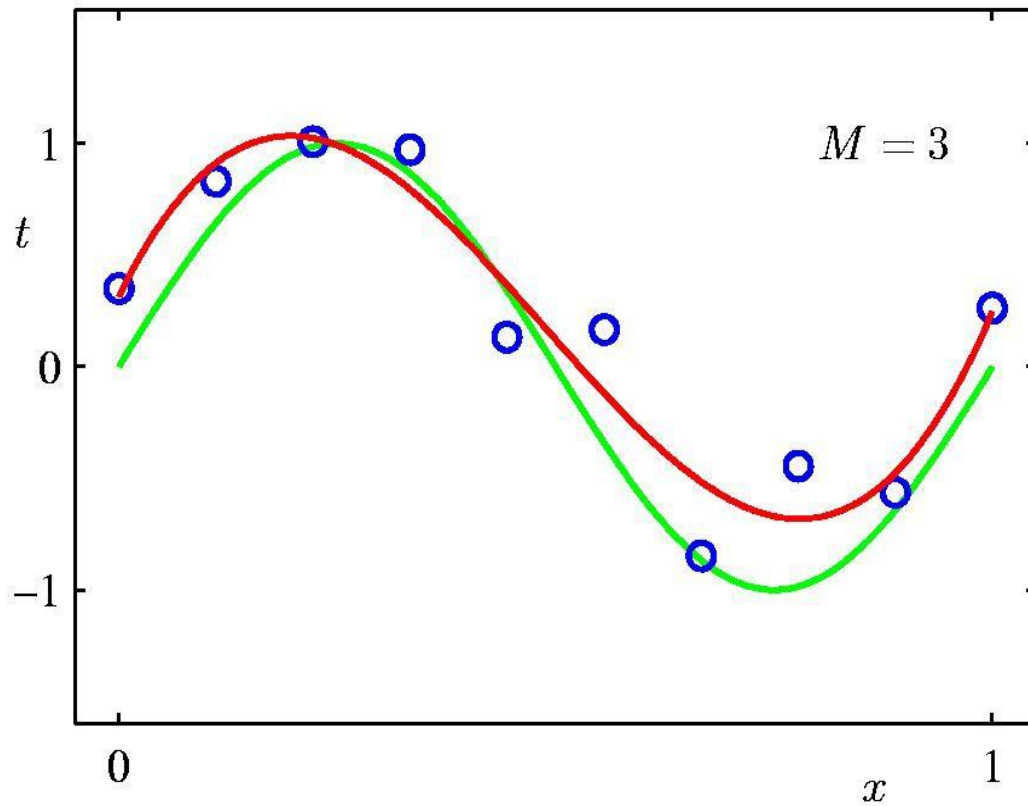
# 0<sup>th</sup> Order Polynomial



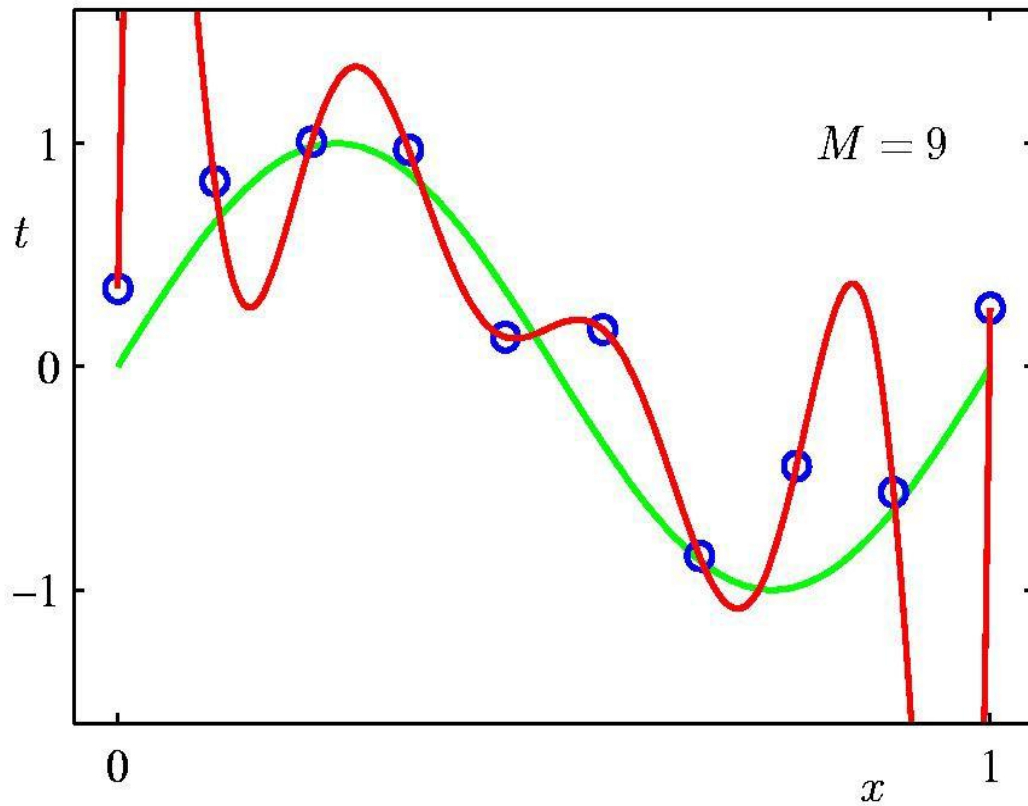
# 1<sup>st</sup> Order Polynomial



# 3<sup>rd</sup> Order Polynomial



# 9<sup>th</sup> Order Polynomial



# Three Types of Error

- Given a **training set** of size  $N$ , we can measure the model performance based on the **training error**: e.g.

$$ASE = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2$$

Other forms of errors could be considered.

- We are more interested in the **true error**, i.e.

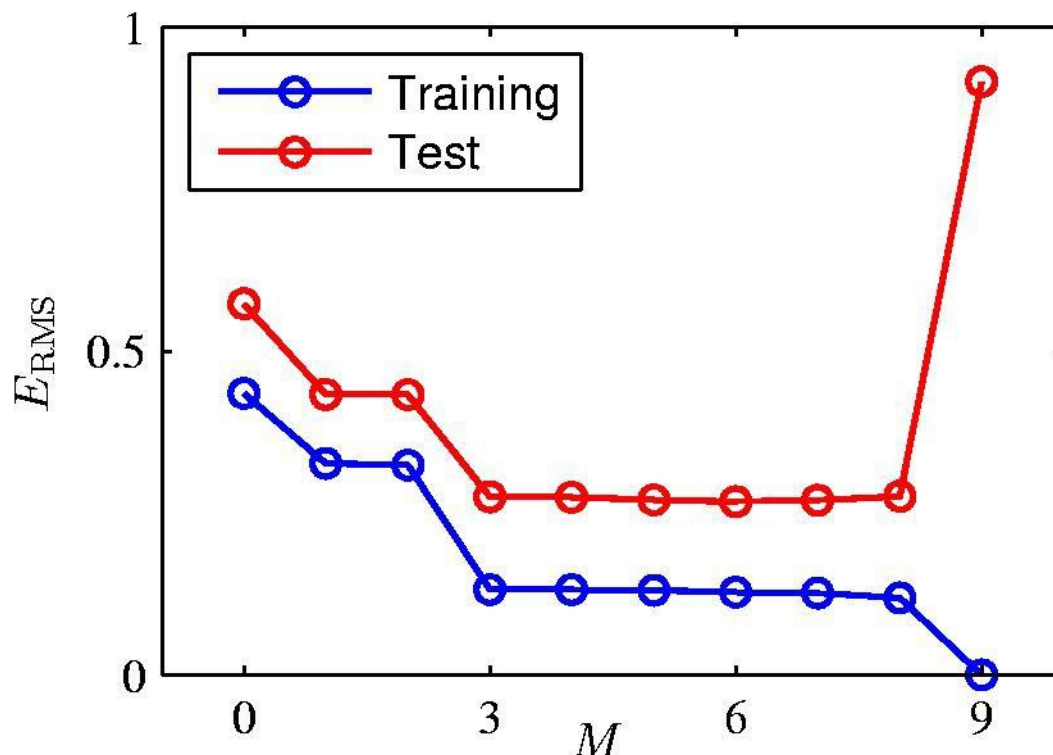
$$MSE = E[(y_n - \hat{y}_n)^2]$$

Estimate it based on a **test set** (not used in training)

**Testing error:**

$$ASE = \frac{1}{|TestSet|} \sum_{n \in TestSet} (y_n - \hat{y}_n)^2$$

# Over-fitting



Root-Mean-Square (RMS) Error:  $E_{\text{RMS}} = \sqrt{2E(\mathbf{w}^*)/N}$

Low training error does not always implies low testing error

# Data Mining Emphasizes ...

- Techniques **feasible** to be used for **large** datasets
- Possibility of **generalizing results**
  - **Patterns** found in **training** dataset still appear in **new** dataset
  - **Predictive** performance
- **Actionable** results
  - To suggest appropriate **strategies / policy / prediction**

# Unsupervised Learning

## ■ Objectives:

- Understand **structure** in data
- Find well-separated **groups** of **cases**, or form **prototypes**

## ■ No target variable

- No **“teacher”** to provide guidance
- Assess based on **practical value** of discovered structure

## ■ Useful when there is **redundancy** in the data

|       |     | Feature/Input Variables |       |       |       |     |       |
|-------|-----|-------------------------|-------|-------|-------|-----|-------|
|       |     | $X_1$                   | $X_2$ | $X_3$ | $X_4$ | ... | $X_r$ |
| Cases | 1   |                         |       |       |       | ... |       |
|       | 2   |                         |       |       |       | ... |       |
|       | 3   |                         |       |       |       | ... |       |
|       | 4   |                         |       |       |       | ... |       |
|       | 5   |                         |       |       |       | ... |       |
|       | ⋮   | ⋮                       | ⋮     | ⋮     | ⋮     |     | ⋮     |
|       | ⋮   | ⋮                       | ⋮     | ⋮     | ⋮     |     | ⋮     |
|       | $n$ |                         |       |       |       | ... |       |

# Unsupervised Learning in scikit-learn

## 2. Unsupervised learning

- ▶ 2.1. Gaussian mixture models
- ▶ 2.2. Manifold learning
- ▶ 2.3. Clustering
- ▶ 2.4. Biclustering
- ▶ 2.5. Decomposing signals in components (matrix factorization problems)
- ▶ 2.6. Covariance estimation
- ▶ 2.7. Novelty and Outlier Detection
- ▶ 2.8. Density Estimation
- ▶ 2.9. Neural network models (unsupervised)

# Association Rules



transaction id



| <i><b>TID</b></i> | <i><b>Items</b></i>       |
|-------------------|---------------------------|
| 1                 | Bread, Coke, Milk         |
| 2                 | Beer, Bread               |
| 3                 | Beer, Coke, Diaper, Milk  |
| 4                 | Beer, Bread, Diaper, Milk |
| 5                 | Coke, Diaper, Milk        |

## ■ Data:

- A set of transactions, each of which consists of a set of items

## ■ Association rules

- A set of rules that characterize associations between items
- Eg. 90% of people who purchased items A and B also purchased item C

E.g., Customers who bought this item also bought

Amazon.com: Harry Potter and the Sorcerer's Stone (Book 1): Books: J.K. Rowling, Mary GrandPré - Microsoft Internet Explorer

檔案(F) 編輯(E) 檢視(V) 我的最愛(A) 工具(T) 說明(H)

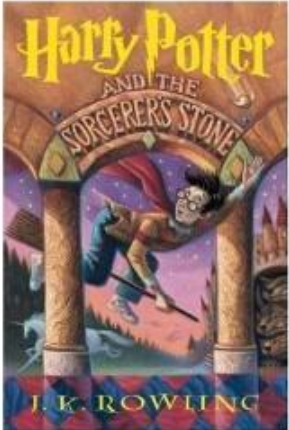
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**Harry Potter and the Sorcerer's Stone (Book 1) (Hardcover)**  
by J.K. Rowling (Author), Mary GrandPré (Illustrator)  
★★★★★ (5,397 customer reviews)

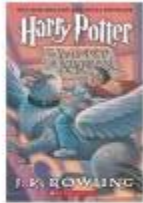
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Price: **\$15.63** & eligible for **FREE Super Saver Shipping** on orders over \$25. [Details](#)  
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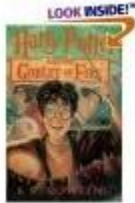
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**Harry Potter and the Prisoner of Azkaban (...)** by J.K. Rowling  
★★★★★ (2,585) \$15.63



**Harry Potter and the Goblet of Fire (Book 4)** by J.K. Rowling  
★★★★★ (5,158) \$19.79



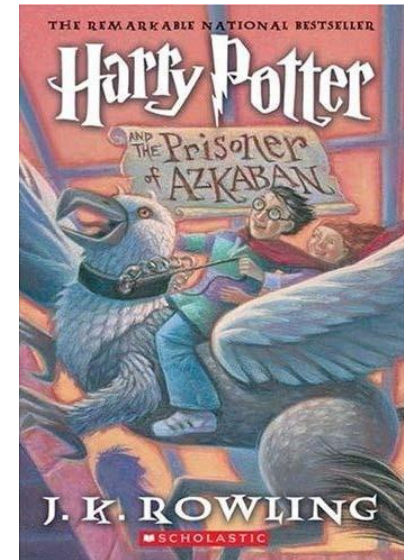
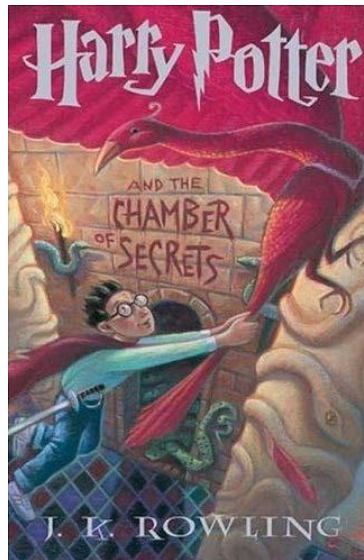
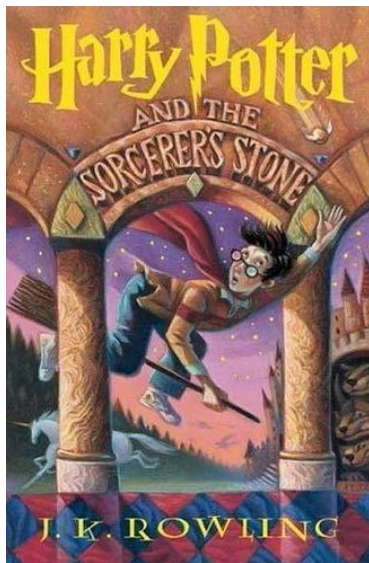
**Harry Potter and the Order of the Phoenix...** by J. K. Rowling  
★★★★★ (5,874) \$9.99



**Harry Blood...** by J.K. Rowling  
★★★★★

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# Sequential Patterns



E.g., active promotion, disease, web page traversal pattern, alarm, ...

# Example 1: Customer Attrition / Churn Analysis in Telcos

- **Goal:** To prevent loss of customers (i.e. to predict whether a customer is likely to be lost to a competitor)
- **Data:** telephone usage of customers
- **Methods:** use neural network, time series analysis to identify typical patterns of likely-to-churn customers
- **Benefit:** Retention of customers, more effective promotions.

# Example 2: Fraud Detection in Automobile Insurance

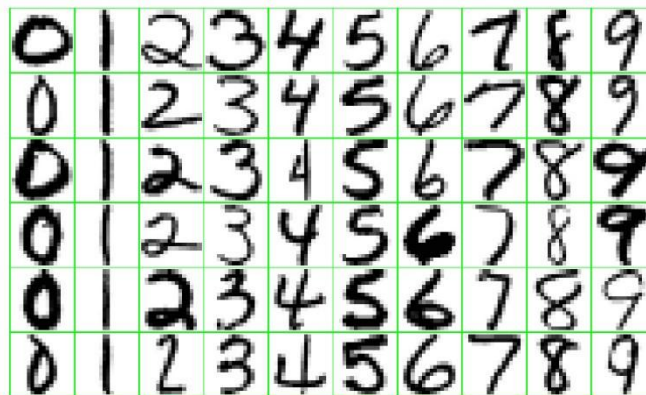
- **Goal:** fraud increases costs or reduce revenue.
- **Data:** past reports on claims investigated by experts to identify cases of fraud.
- **Methods:** use logistic regression to identify characteristics of fraudulent cases to prevent in future
- **Benefit:** Increase profits by reducing undesirable customers.

# Example 3: Credit scoring in Bank loans

- **Goal:** reduce risk of loans to delinquent customers.
- **Data:** past loan application results.
- **Methods:** use credit scoring models based on logistic regression, discriminant analysis to create score functions that separate out risky customers.
- **Benefit:** Decrease in cost of bad debts.

# Example 4: Handwritten Digit Recognition

- **Goal:** identify single digits 0 – 9 based on handwritten ZIP codes on envelopes.
- **Raw data:** images that are scaled segments from five digit ZIP codes.
  - 16x16 eight-bit grayscale maps
  - Pixel intensities range from 0 (black) to 255 (white)
- **Input data:** 7291 digits (256 dimensions each)



# Information categorization approach to literary authorship disputes

Albert C.-C. Yang<sup>a,b,c,\*</sup>, C.-K. Peng<sup>a</sup>, H.-W. Yien<sup>b,c</sup>,  
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Beth Israel Deaconess Medical Center/Harvard Medical School, Boston, MA 02215, USA*

<sup>b</sup>*School of Medicine, National Yang-Ming University, Taipei, Taiwan*

<sup>c</sup>*Taipei Veterans General Hospital, Taipei, Taiwan*



## 程甲本《紅樓夢》程偉元序

《紅樓夢》小說本名《石頭記》，作者相傳不一，究未知出自何人，惟書內記曹雪芹先生刪改數過。好事者每傳抄一部，置廟市中，昂其值，得數十金，可謂不脛而走者矣。然原目一百廿卷，今所傳只八十卷，殊非全本。即間稱有全部者，及檢閱，仍只八十卷，讀者頗以為憾。不佞以是書既有百廿卷之目，豈無全璧？爰為竭力收羅，自藏書家甚至故紙堆中無不留心，數年以來，僅積有廿余卷。一日偶於鼓擔上得十餘卷，遂重價購之，欣然繙閱，見其前後起伏，尚屬接筭，然漶漫不可收拾。乃同友人細加厘剔，截長補短，抄成全部，復為鐫板，以公同好，《紅樓夢》全書始自是告成矣。書成，因並志其緣起，以告海內君子。凡我同人，或亦先睹為快者歟？小泉程偉元識

| Rank | 1-40 |           | 41-80 |           | 81-120 |           |
|------|------|-----------|-------|-----------|--------|-----------|
|      | Word | Frequency | Word  | Frequency | Word   | Frequency |
| 1    | 了    | 6250      | 了     | 8301      | 了      | 6946      |
| 2    | 不    | 4505      | 不     | 5676      | 的      | 5499      |
| 3    | 的    | 4010      | 的     | 5539      | 不      | 5009      |
| 4    | 一    | 3891      | 一     | 4942      | 來      | 3944      |
| 5    | 道    | 3683      | 來     | 4097      | 道      | 3756      |
| 6    | 來    | 3563      | 人     | 3892      | 是      | 3741      |
| 7    | 人    | 3139      | 我     | 3769      | 人      | 3644      |
| 8    | 我    | 2843      | 是     | 3720      | 一      | 3461      |
| 9    | 是    | 2833      | 道     | 3683      | 說      | 3391      |
| 10   | 說    | 2805      | 說     | 3637      | 我      | 2743      |

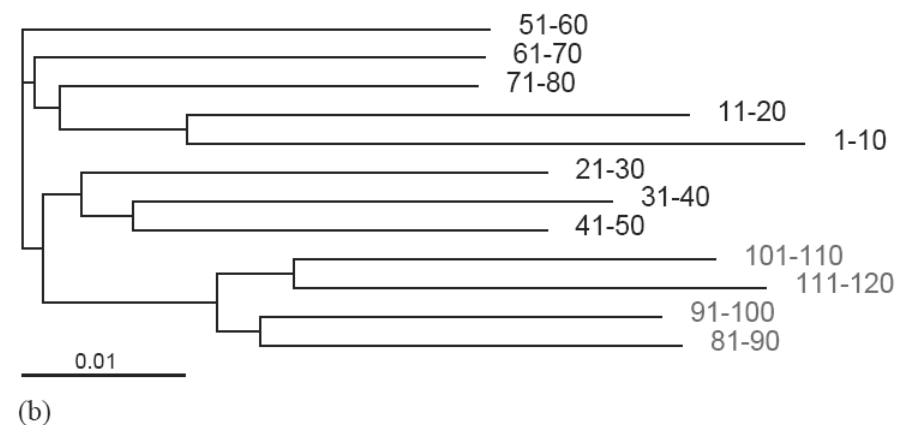
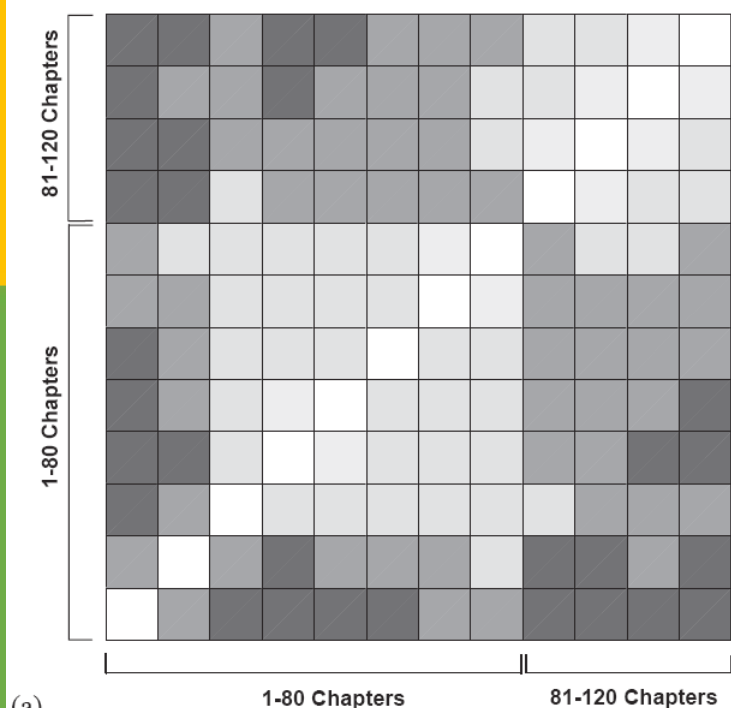


Fig. 3. Rank frequency analysis of *The Dream of the Red Chamber*. (a) Pairwise distance matrix of the entire book. Each box represents a segment containing 10 consecutive chapters. The distance between each pair of segments is represented on a graded color scale. (b) The neighbor-joining phylogenetic tree constructed by the pairwise distance matrix. The distance between each segment on the tree is the summation of the total horizontal lengths along the connection path. Both representations support the view that the first 80 chapters and last 40 chapters are by different authors.

# Data Mining Software

- Many, see [www.kdnuggets.com](http://www.kdnuggets.com)
- Major commercial software
  - SAS Enterprise Miner [www.sas.com](http://www.sas.com)
  - IBM SPSS Modeler [www.spss.com](http://www.spss.com)
  - Salford Systems Predictive Modeling [www.salford-systems.com/](http://www.salford-systems.com/)
  - Frontline Systems XLMiner [www.solver.com/xlminer-data-mining/](http://www.solver.com/xlminer-data-mining/)
- Free software
  - Python (Anaconda <https://www.anaconda.com/download/>)
  - Weka [www.cs.waikato.ac.nz/ml/weka/](http://www.cs.waikato.ac.nz/ml/weka/)
  - RapidMiner [rapidminer.com](http://rapidminer.com)
  - Rattle with R [rattle.togaware.com/](http://rattle.togaware.com/) [www.r-project.org/](http://www.r-project.org/)

# Python

- Open source
- Python is TIOBE (software quality Co)'s programming language of the year 2018!
- Interpreted language
- Recommended plotting library: matplotlib
- **Specialties**
  - Strong text processing
  - Simple to use
  - Support for object oriented programming

# R

- Open source
- Based on S language for statisticians
- Popular language for data analysis
- Recommended plotting library: ggplot2
- **Specialties**
  - Vector/matrix operations
  - But loops are slow
  - Table data structure (data frames)

# Topics to be Covered

- Feature Engineering and Selection
- Supervised Learning
  - Decision Tree
  - Ensemble Methods
  - Support Vector Machines
  - Naive Bayes
  - Neural Networks and Intro to Deep Learning
- Unsupervised Learning
  - Cluster Analysis
  - Association Rules
- Text Mining

# Summary:

## Keys for Successful Data Mining

- Do you have clear objectives?
  - Are the relevant data available?
  - Are the data of good quality?
  - Have you missed any relevant qualitative information?
    - Wall Street Journal (March 21, 2012): Lululemon's sales strategy: scarcity, eavesdropping
- <https://www.youtube.com/watch?v=2mkxMQGG2z4>
- Are you able to generate actions from the mining results?

